

The CloudSim Framework: Modelling and Simulating the Cloud Environment

CloudSim provides a generalised and extensible simulation framework that enables seamless modelling and simulation of app performance. By using CloudSim, developers can focus on specific systems design issues that they want to investigate, without getting concerned about details related to cloud-based infrastructures and services.



Advances in computing have opened up many possibilities. Hitherto, the main concern of application developers was the deployment and hosting of applications, keeping in mind the acquisition of resources with a fixed capacity to handle the expected traffic due to the demand for the application, as well as the installation, configuration and maintenance of the whole supporting stack. With the advent of the cloud, application deployment and hosting has become flexible, easier and less costly because of the *pay-per-use* chargeback model offered by cloud service providers.

Cloud computing is a best-fit for applications where users have heterogeneous, dynamic, and competing quality of service (QoS) requirements. Different applications have different performance levels, workloads and dynamic application scaling requirements, but these characteristics, service models and deployment models create a vague situation when we use the cloud to host applications. The cloud creates complex

provisioning, deployment, and configuration requirements.

Why simulation is important for the cloud environment

Cloud service providers offer elastic, on-demand, and measured infrastructure, platforms and software services. In the public cloud, tenants have control over the OS, storage and deployed applications. Resources are provisioned in different geographic regions. In the public cloud deployment model, the performance of an application deployed in multiple regions is a matter of concern for organisations. Proof of concepts in the public cloud environment give a better understanding, but cost a lot in terms of capacity building and resource usage even in the *pay-per-use* model.

CloudSim, which is '...a toolkit for the modelling and simulation of Cloud computing environments' comes to the rescue. It provides system and behavioural modelling of the Cloud computing components. Simulation of cloud